

Towards Better
Running Time

Let's focus on Run time.

Ford - Fulkerson - limitation

- pseudo polynomial

Poly. - if all edge Capacities are polynomially bounded integers

with large edge Capacities

- if could take many iterations

Root Cause

- Repeatedly choose "bad" augmenting path

- path selection is arbitrary

- Leads to Slow
Convergence

How to fix this?

Idea:- * Choose Augmenting
paths Carefully.

* Avoid repeatedly
making same
mistakes.

Refinement:- Edmond - Karp

Always select

- shortest augmenti-
-ng path.

How to find shortest path?

Use BFS / DFS
(could also use Dijkstra, but overkill)

Edmonds-Karp

initialize $f_e = 0$, $\forall e \in E$.

repeat

Compute an $s \sim t$ path
(with ~~the~~ residual
Capacity) in the
current G_f with
fewest # of edges

1) Takes $O(|E|)$ by
BFS.

if no such path then
halt with current
flow $\{f_e\}_{e \in E}$.

else

let $\Delta = \min_{e \in P} (\text{e's residual capacity in } G_f)$

1) Augment the
flow f using the path
 P .

for all edges e of G whose
corresponding forward edge
is in P do
increase f_e by Δ

for all edges e of G whose
 correspondingly reverse edge
 is in P do
 decrease f_e by Δ

Analysis

Correctness we can inherit
 from Ford - Fulkerson.

Running time

$$- O(m^2 n)$$

$$m \text{ --- } \lesssim n \text{ (sparse)}$$

$$\text{--- } \lesssim n^2 \text{ (dense)}$$

So, betⁿ n^3 & n^5 .

E-K progress lemma

Fix a network G .

For a flow f , let $d(f)$ denote # of hops in shortest $s \rightarrow t$ path (with the residual capacity) in G_f , or $+\infty$ if no such paths exist

(a) $d(f)$ never decreases during the execution of the E-K.

(b) $d(f)$ increases at least once per m iterations.

Obs. about dist.

$$d(t) \in \{0, 1, 2, \dots, n-1, +\infty\}$$

$$\text{if } d(t) \geq n$$

$$\text{— Then } d(t) = +\infty$$

$\Rightarrow$ No path from s to t in residual graph G_f

— Alg. performs $O(m \times n)$ iterations.

— Each iteration — run BFS.

$$\text{Total — } O(m^2 n)$$

BFS layer structure

It partitions into layers

- layer $-0-$ S .
- layer $i+1$: vertices reachable in one step from layer i

Edge classification in G_f

($\longrightarrow$) Forward edges
(i)th layer $\rightarrow$ ($i+1$)th layer

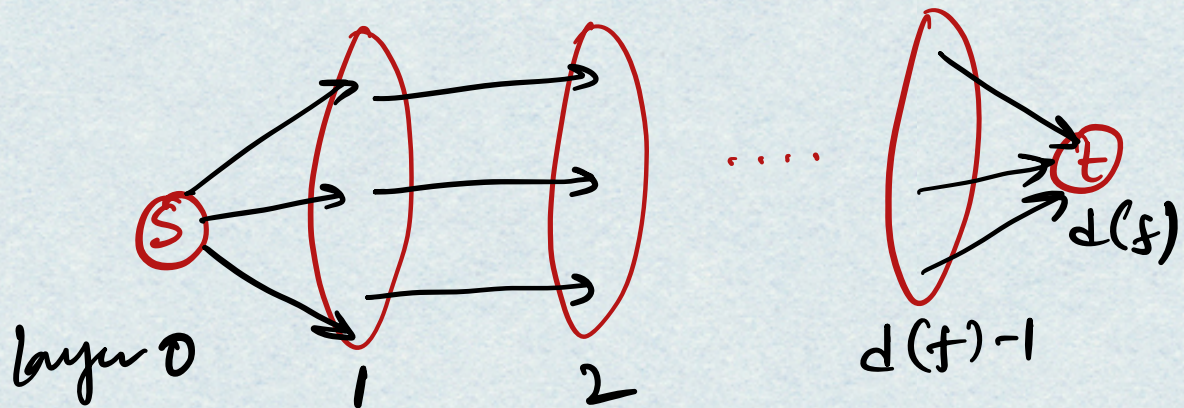
Sideways edges

- in the same layer

($\longleftarrow$) Backward edges

From layer $i \rightarrow j$,
where $j < i$

Define G_f



This is a succinct
encoding of all the
shortest paths of G_f

paths on which
Edmond-Karp might
augment!

Every $s \sim t$ path in G_f comprises of only forward edges of the BFS, Hence

$d(f)$ many hops.
min. possible

Conversely,

an $s \sim t$ path that is in G_f but not in L_f must contain at least one detour

Hence, requires $d(f) + 1$ hops to get to t

